

BHABHA UNIVERSITY, BHOPAL

NH-12, Hoshangabad Road, Jatkhedhi, Bhopal-462026

MASTER IN COMPUTER APPLICATION (MCA)

First Semester

Scheme of Examination



DETAILED SYLLABUS (EFFECTIVE FROM JULY 2018)

**Department of Computer Science & Application
BHABHA UNIVERSITY, BHOPAL**

MASTER IN COMPUTER APPLICATION (MCA)

First Semester

Scheme of Examination

S.No	Subject Code	Subject Name	Periods per week			Credits	Maximum Marks (Theory slot)			Maximum Marks (Practical Slot)		Total Marks
			L	T	P		End Sem Exam	Tests	Assignment / Quiz	End Sem Practical Viva	Practical Record/Assignment/Quiz/Presentation	
1	MCA11	Computer Fundamental	3	1	-	4	70	20	10		-	100
2	MCA12	Computer Organization and Architecture	3	1	-	4	70	20	10	-	-	100
3	MCA13	Problem solving in c language	3	1	-	4	70	20	10	-	-	100
4	MCA14	Discrete Mathematical Structure	3	1	-	4	70	20	10	-	-	100
5	MCA 15	Business English & Communication	3	1	-	4	70	20	10	-	-	100
6	MCA 16	Programming Lab In C	-	-	8	8	-	-	-	100	50	150
7	MCA17	Microprogramming and Architecture Lab	-	-	2	2	-	-	-	30	20	50
		Total	15	5	10	30	350	100	50	130	70	700

MCA-11 Computer Fundamental

UNIT-I

Computer Concept, Computer System Characteristics, , Basic Components of A Computer System Basic concepts of IT, concepts of Data & Info, data processing, history of computers (generation, type of languages), organization of computers, I/O devices, storage devices, system software, application software, utility packages, numerical based on storage devices.

UNIT-II

Assembler : Elements of assembly language programming, a simple assembly scheme, pass structure of assembler, design of two pass assemblers, a single pass assemblers., Primary Vs Secondary, Various Storage Devices - Magnetic Tape, Magnetic Disks, Floppy Disks, Optical Disks, Pen Drive.

UNIT-III

Need, Types of Software - System Software, Application Software, System Software - Operating System, Utility Program, Programming Languages - Machine, Assembly, High Level, 4GLs., Assemblers, Compilers and Interpreter, Application Software – Word Processing, Spreadsheet, Presentation Graphics, Data Base Management Software.

UNIT-IV

Various Applications of Computer, Programming Techniques Various Stages of Program Development . Algorithms, Flow Charts - Symbols, Rules for Making Flow Chart, Types of Flow-Chart, Advantage & Disadvantage, Pseudo Codes, Programming , Various Looping Statements, Various Branching Statements, Decision Tree, Decision Table, Dry Running of Program.

UNIT-V

Sequential file organization, random file organization, index structure, indexed file organization, alternate key indexed sequential files, multi key organization, multi key access, multi list file organization, inverted files & their definitions, insertion, deletion, operations with optimum utilization of memory, comparison of various type of file organization.

BOOKS:

1. D.M. Dhamdhere “ System Programming & O.S.” 2nd Ed., Tata Mc. Graw Hill. J. Donovan “System Programming” THM.
2. COMPUTERS TODAY BY S.K.BASANDRA, GALGOTIA PUBLICATIONS.
3. Fundamentals of information technology by Alexis leon & mathews leon,
4. MS-office complete reference bpb publication

MCA-12 Computer Organization and Architecture

Unit-I

Data representation Data types and Number systems, Binary Number System, Octal and Hexadecimal Number system, Fixed point Representation, 1's & 2's complement, Binary Fixed point Representation, Arithmetic operation on Binary Numbers, Overflow and Under flow, Floating point representation, ASCII Codes, EBCDIC codes, Gray Code, Excess-3 & BCD, Error detection & correction codes

Unit-II

Boolean algebra and digital logic circuits- Logic gates, AND, OR, NOT Gates and their truth tables, NOR, NAND & XOR Gates, Boolean Algebra, Basic Definition and properties, Basic Boolean Law's, Demorgan's Law, K Map-, SOP & POS, Don't care conditions

Unit-III

Combination Circuits-Half adder, Full adder, Full subtractor and decimal adder, Multi level NAND and NOR Circuits, decoders, Multiplexers and Demultiplexers.

Unit-IV

Sequential logic-Flip-Flops-RS, D, JK & T Flip-Flop, Triggering in Flip flops, flip flop excitation tables, Design Procedure and design of counters

Unit-V

Memory organization: Secondary memory, Primary memory: RAM, ROM basic cell of static and dynamic RAM, concept of segmentation and paging, Associative memory, cache memory organization, virtual memory organization

Books

3. Digital Logic and Computer Design By Morris Mano
2. Computer System Architecture By Morris Mano
3. Digital computer Electronics By Malvino

MCA-13 Programming and Problem Solving in C

Unit-I

An overview: Problem identification, analysis, design, coding, testing & debugging, implementation, modification & maintenance; algorithms & flowcharts; Characteristics of a good program - accuracy, simplicity, robustness, portability, minimum resource & time requirement, modularization; Rules/ conventions of coding, documentation, naming variables; Top down design; Bottom-up design.

Unit-II

Fundamentals of C Programming: History of C; Structure of a C Program; Data types; Constant & Variable, naming variables; Operators & expressions; Control Constructs – if-else, for, while, do-while; Case switch statement; Arrays; Formatted & unformatted I/O; Type modifiers & storage classes; Ternary operator; Type conversion & type casting; Priority & associativity of operators.

Unit-III

Modular Programming: Functions; Arguments; Return value; Parameter passing – call by value, call by reference; Return statement; Scope, visibility and life-time rules for various types of variable, static variable; Calling a function; Recursion – basics, comparison with iteration, types of recursion- direct, indirect, tree and tail recursion, when to avoid recursion, examples.

Unit-IV

Advanced Programming Techniques: Special constructs – Break, continue, exit(), goto & labels; Pointers - & and * operators, pointer expression, pointer arithmetic, dynamic memory management functions like malloc(), calloc(), free(); String; Pointer v/s array; Pointer to pointer; Array of pointer & its limitation; Function returning pointers; Pointer to function, Function as parameter; Structure – basic, declaration, membership operator, pointer to structure, referential operator, self referential structures, structure within structure, array in structure, array of structures; Union – basic, declaration; Enumerated data type; Typedef; command line arguments.

Unit-V

Miscellaneous Features: File handling and related functions; printf & scanf family; C preprocessor – basics, Include, #define, undef, conditional compilation directive like if, else, elif, endif, fdef and ifndef; Variable argument list functions.

BOOKS:

1. Kernighan & Ritchie “The C programming language”, PHI
2. Schildt “C: The Complete reference” 4th ed TMH.
3. Cooper Mullish “The Spirit of C”, Jaico Publishing House, Delhi
4. Kanetkar Y. “Let us C”, BPB.
5. Kanetkar Y.: “Pointers in C”, BPB
6. Gottfried : “Problem Solving in C”, Schaum Series
7. Jones, Harrow Brooklish “C Programming with Problem Solving”, Wiley Dreamtech India.

MCA-14 Discrete Mathematical Structure

Unit-I

Sets, Relations and Functions: Sets, Subsets, Power Sets, Complement, Union and Intersection, D Morgan's Law, Cartesian Products Relations, relational matrices, Properties of Relations, Equivalence relation, Functions, Injection, Surjection and Injective mapping composition of functions and Mathematical Induction.

Unit-II

Lattices & Boolean Algebra: Lattices: Introduction, Partial ordered sets, combination of partial ordered sets, Hasse diagram, Introduction of Lattices, properties of lattices-Bounded, Complemented, Modular and complete lattice **Boolean algebra:** Introduction, Axioms and Theorems of Boolean algebra, Boolean functions simplification of Boolean functions and Boolean algebra.

Unit-III

Propositional & Predicate logic: Propositions, Truth tables Tautology, Contradiction, Algebra of propositions Theory of Inference and natural deduction, Theory of predicates, first order predicate, predicate formulas Quantifiers, inference theory of predicate logic.

Unit-IV

Trees & Graphs: Finite graphs, incidence and degree, isomorphism, sub graphs and union of graphs, connectedness, walk paths and circuits Eulerian graphs, tree properties of trees, pendent vertices in tree centre of tree, spanning trees and cut vertices, binary trees, matrix representation of graph, incidence and Adjacency matrix and their properties, applications of graph in computer science.

Unit-V

Discrete Numeric function and Recurrence Relation: Introduction to discrete numeric functions and generating functions introduction to recurrence relations and Recursive algorithms, linear recurrence relations with constant coefficients, homogeneous solutions, particular Solutions and total solutions.

Books

- 1 J.P. Trembley & R.P. Manohar "Discrete Mathematical Structure with applications to computer Science"
- 2 Kenneth H Rosen-203 "Discrete Math & Its Applications" 5^{ed}
- 3 Bernard Kolman & Robart C. Busby "Discrete Mathematical Structure for computer Science"

MCA-15 Communication Skills

Unit-I

Communication: Meaning and process of communication, importance of effective communication, communication situation, barriers to communication, objectives of communication, types of communication, principles Of communication essentials of effective communication.

Unit-II

Media of communication: Written, oral, face to face, visual, audio-visual, merits and demerits of written and oral communication.

Unit-III

Communication Skills: Developing Communication Skills, Listening, Speaking, Reading-writing (Oral & Written) Body language, Utility of aids in communication.

Unit-IV

Spoken Skills: Preparing for oral presentation, conducting presentation, Debates, Seminar, speeches, Lectures Interviews, Telephonic Conversation, Group Discussions.

Unit-V

Written Skills: Preparing of Bio-Data, seminar, paper, bibliography and official correspondence, Mechanics Of writing Formal and Informal writings, letters, paragraphing, precise ,report writing, technical reports, length Of written reports, organizing reports, writing technical reports, creative writing, common errors in Language

Books

4. Rajendra Pal and J.S. Korlahalli “Essentials of Business communication”, Sultan chand and sons publishers, New Delhi.
- 2 U.S. Rai and S.M. Rai “Business Communication” Himalaya publishing House
- 3 Menzal and D.H. Jones “writing a technical paper” Mc Graw Hill 1961